

TOOLS FOR TROUBLESHOOTING THEORY

1.List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

ANS:

Real-Life Scenario	How Multimeter Helps
1.Computer power supply not working	Checks the SMPS output voltage to confirm whether it is supplying the correct power.
2.Laptop/phone charger not charging	Measures the charger’s output voltage to find out if the adapter is working properly.
3.Smartphone battery or charging circuit problem	Measures battery voltage and checks for continuity to identify a faulty battery or connection.

2.Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

ANS:

Brief Summary

The POST card displays hexadecimal codes during startup to show which stage of the hardware **Power-On Self-Test (POST)** is running.
In the demonstrated example, codes such as **3387, 2900, and AA00** appeared, indicating BIOS checking and then successful booting.
The codes help technicians identify where startup stops, such as problems with **CPU, RAM, GPU, or motherboard**.
In that video, the POST card helped confirm that the system completed its checks and successfully started the operating system.

3.Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

ANS:

Procedure:

1. Download and install HWMonitor.
2. Open HWMonitor and allow it to detect the hardware.
3. Check the CPU/GPU temperature, voltage, and fan-speed readings.

4. Take a screenshot of the main dashboard.
5. Crop or blur any personal information before submitting.

Result:

HWMonitor successfully displayed the system's hardware health details, helping to monitor temperature, voltage, and other hardware conditions.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

ANS:

Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, resistance, and continuity in electrical circuits.	Faulty SMPS power output or damaged charging cable.
POST Card	Checks the computer's startup/POST process using diagnostic codes.	RAM, CPU, or motherboard failure preventing boot.
Software Diagnostics	Checks hardware status, temperatures, voltages, and drive health through software.	Overheating CPU or a failing SSD/HDD.